N-of-1 Longitudinal Study of a Patient with ME/CFS: Methods and Preliminary Results

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Abstract

Keywords: n-of-1; intensive longitudinal data; ME/CFS; fNIRS; EEG; HRV; POTS

1 Introduction

2 Methods

2.1 Study design

2.2 Data acquisition

2.2.1 Symptom scores

2.2.2 Physical activity

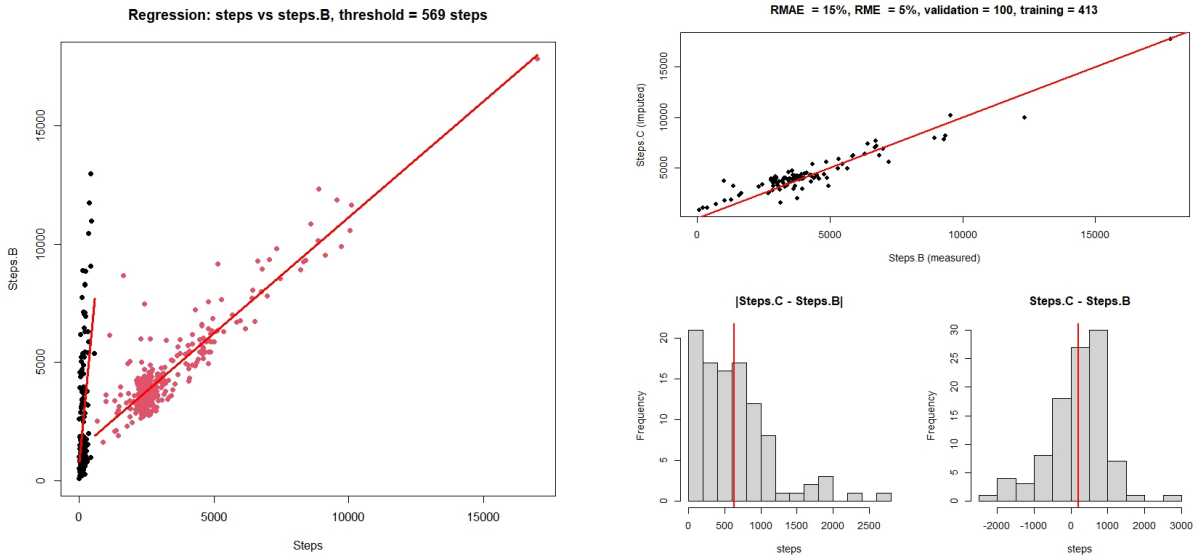
Daily step counts were recorded with a smartphone (Redmi Note 11S/Pedometer Step Counter) from January 2023, and with a wrist-worn watch (Dyurawa) from March 2025. The phone records only the steps taken while it is carried, so it misses most activity at home, whereas the watch records activity throughout the day; watch counts were therefore taken as the reference measure.

Gaps of up to four consecutive days were filled by linear interpolation in both series, if missing days were less than 10%. Days on which the watch recorded fewer steps than the phone were considered days of watch non-wear or malfunction, and their watch counts were set to missing.

To obtain a continuous series of watch-equivalent steps over the whole study period (`Steps.C`), watch counts were estimated from phone counts on days without watch data. The relationship between the two counts differs between days spent at home, when the phone is mostly not carried and its count carries little information, and days with outdoor activity, when the phone is carried. Days were therefore divided into two regimes by a threshold t on phone steps (`Steps`), and a linear regression of watch steps (`Steps.B`) on phone steps was fitted separately in each regime. The threshold was chosen by scanning values from 50 to 2000 phone steps and selecting the one that minimised the total sum of squared residuals of the two regressions, requiring at least 20

days in each regime (see Figure 1a). On days with watch data, `Steps.C` equals the measured watch count; on the other days, it is the value predicted by the regression of the corresponding regime.

The imputation was validated by fitting the model on the first 80% of the watch period and comparing the values predicted for the remaining 20% with the measured ones (Figure 1b). The relative mean absolute error, $\sum |\hat{y}_i - y_i| / \sum y_i$, and the relative mean error (bias), $\sum(\hat{y}_i - y_i) / \sum y_i$, are indicated on the plot. Most days were slightly overestimated, while the most active days, and outdoor activity without the phone, were underestimated. The time series of `Steps`, `Steps.B`, and `Steps.C` are shown in Figure 2.



(a) Regression model. (b) Validation.

Figure 1: Imputation of watch steps (`Steps.B`) from phone steps (`Steps`).

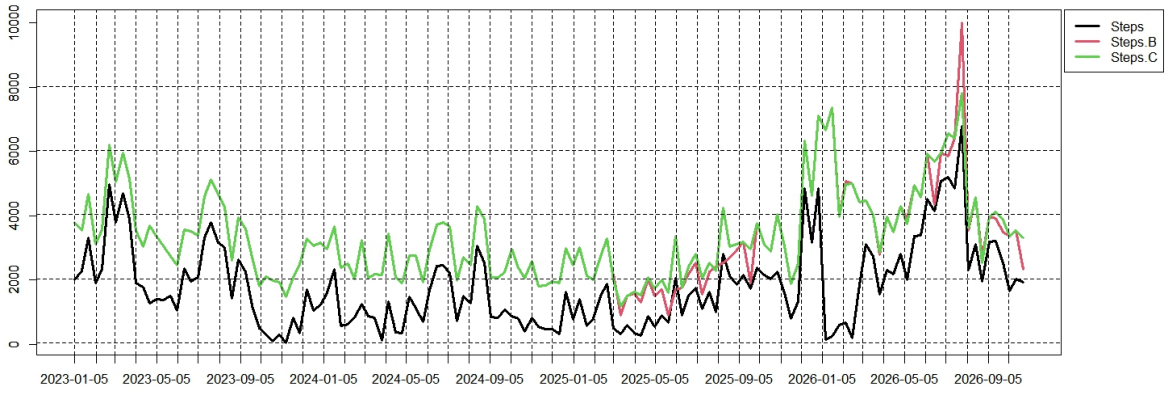


Figure 2: Daily steps averaged over non-overlapping 10-day blocks.

2.2.3 Heart rate variability

Heart rate variability (HRV) was assessed from five-minute recordings of RR intervals, following the protocol and the analysis pipeline described in detail in a previous work [1], to which the reader is referred for an overview on HRV, the definition of the indices, their physiological interpretation, and an introduction to spectral analysis by Discrete Fourier Transform. Recordings started on 2022-07-09 and, up to 2026-10-23, were available for 1438 days, and were performed once a day, simultaneously with the EEG and fNIRS recordings. They were performed once a day, in the morning, in the supine position, after five minutes of rest in the same position. RR intervals were acquired for about five minutes with a Polar H9 sensor mounted on a moistened chest belt, a type of device whose agreement with the electrocardiogram has been documented [2]. The sensor was connected via Bluetooth to a smartphone running the Elite HRV application, and the RR series were exported in text format, one file per recording.

The series were analysed with the R package RHRV [3]. For each recording, the instantaneous non-interpolated heart rate was computed from the RR intervals, and anomalous beats (ectopic beats and artefacts) were removed with the adaptive filter of the package, which discards beats that differ too much from the preceding ones or that fall outside physiological limits (default parameters). The filtered series was then linearly interpolated at 4 Hz to obtain the evenly sampled signal required by the discrete Fourier transform. Recordings shorter than 240 s or longer than 360 s were excluded, and when more than one recording was available for the same day only the first one was retained.

In the time domain, the following indices were computed: mean heart rate (HR) and mean RR interval (RR); the standard deviation of the RR intervals (SDNN) and the HRV triangular index (HRVi), as measures of overall variability; the root mean square of successive differences (rMSSD) and the percentage of successive differences greater than 50 ms (pNN50), as measures of the high-frequency, vagally mediated component [3, 4].

In the frequency domain, the power spectrum was estimated with the short-time Fourier transform, using a single window of 300 s, since for five-minute recordings shorter windows give less reliable estimates of the low-frequency power [2]. The power was integrated over the low-frequency (LF, 0.04 Hz to 0.15 Hz) and the high-frequency (HF, 0.15 Hz to 0.40 Hz) bands [5]. The ultra-low- and very-low-frequency bands were not considered, since they cannot be reliably estimated from five-minute recordings [5]. From LF and HF, the following derived quantities were computed: the ratio LF/HF; the total power, defined here as $\mathrm{TotPower} = \mathrm{LF} + \mathrm{HF}$; and the normalised powers $\mathrm{LF}_{\mathrm{nu}} = \mathrm{LF}/(\mathrm{LF} + \mathrm{HF})$ and $\mathrm{HF}_{\mathrm{nu}} = \mathrm{HF}/(\mathrm{LF} + \mathrm{HF})$, which reduce the within- and between-subject variability of the absolute powers [6]. The distributions of the time-domain and frequency-domain indices over all the recordings are shown in Figures 3 and 4.

2.2.4 Blood pressure and lean test

Blood pressure and heart rate were measured with an automatic upper-arm oscillometric monitor (Omron M6 Comfort) immediately after the HRV recording (Section **??**), with the subject still in the supine position, that is after at least ten minutes of supine rest. This measurement provided the supine values of Systolic (SBP) and Diastolic (DBP) blood pressure and of heart rate (HR), and was available on most days of the study.

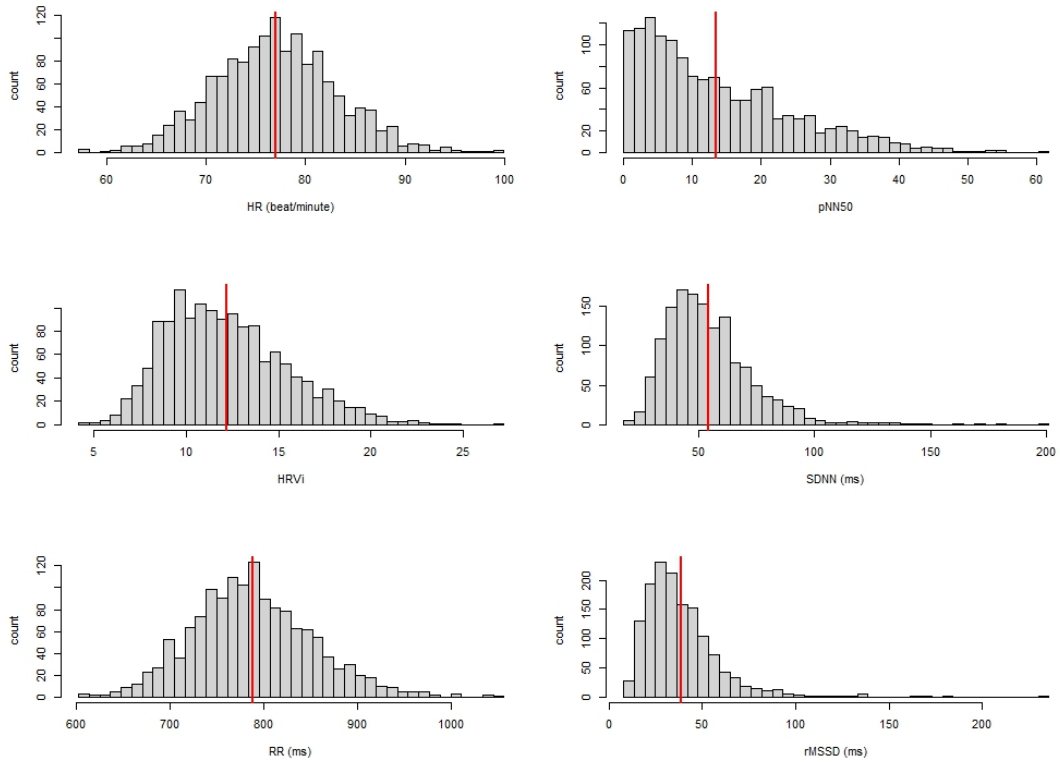


Figure 3: Distributions of the time-domain HRV indices over all the daily five-minute recordings. The mean of each distribution is indicated by a red line.

On some days, the supine measurement was followed by a NASA lean test, performed according to the protocol of the Bateman Horne Center [7], a variant of the passive standing test used by NASA to assess orthostatic intolerance [8]. The subject stood up and leaned against a wall with only the shoulder blades in contact with it and the heels about 15 cm from the wall, keeping still and without talking, a position that reduces the contribution of the leg muscle pump to venous return. Blood pressure and heart rate were then measured during standing, and each measurement was assigned to the minute of standing in which it was taken ($t = 1, 2, \ldots, 10$ min). Unlike the original protocol, which prescribes ten minutes of standing with measurements every minute, the test was performed occasionally, its duration varied between 4 and 10 minutes, and not all time points were recorded in every test; consequently, the later time points are available for fewer tests than the earlier ones. Moreover, the supine values were obtained from a single measurement rather than from two consecutive consistent readings.

For each standing measurement, the changes with respect to the supine values were computed as

$$\Delta\mathrm{HR}_t = \mathrm{HR}_t - \mathrm{HR}_0, \qquad \Delta\mathrm{SBP}_t = \mathrm{SBP}_t - \mathrm{SBP}_0, \qquad \Delta\mathrm{DBP}_t = \mathrm{DBP}_t - \mathrm{DBP}_0, \tag{1}$$

where the subscript 0 denotes the supine measurement. Pulse pressure was defined as $\mathrm{PP} = \mathrm{SBP} - \mathrm{DBP}$ and mean arterial pressure as $\mathrm{MAP} = \mathrm{DBP} + (\mathrm{SBP} - \mathrm{DBP})/3$. Each test was then summarised by the following daily indices: the maximum and the mean of ΔHR, the minimum and the mean of ΔSBP, and the minimum and the mean of PP over the available standing measurements.

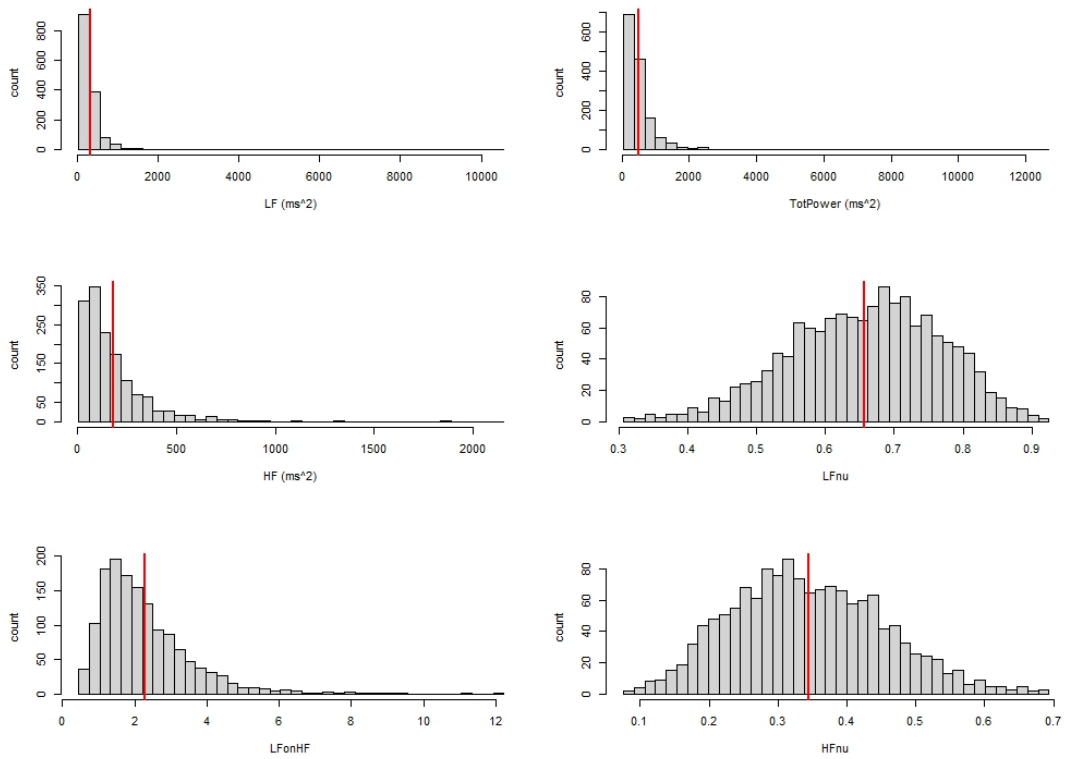


Figure 4: Distributions of the frequency-domain HRV indices over all the daily five-minute recordings. LFonHF is the ratio LF/HF; TotPower is LF + HF. The mean of each distribution is indicated by a red line.

Stroke volume (SV) was not measured directly. An estimate was obtained using the formula proposed by Liljestrand and Zander [9], which expresses SV as a function of Systolic and Diastolic blood pressure:

$$\mathrm{SV} = K\frac{SBP - DBP}{SBP + DBP} \tag{2}$$

where K is a constant to be determined empirically. In a longitudinals study, like the present, the constant K has no effect on the relative changes of SV, which are the quantities of interest. Next, cardiacc output (CO) was estimated as the product of SV and HR, and cardiac power (CP) as the product of CO and MAP:

$$\mathrm{CO} = \mathrm{SV} \cdot \mathrm{HR}, \qquad \mathrm{CP} = \mathrm{CO} \cdot \mathrm{MAP} \tag{3}$$

It may be useful to remeber that SV represents the volume of blood pumped by the left ventricle in one beat, CO represents the volume of blood pumped by the left ventricle in one minute, and CP represents the mechanical power generated by the left ventricle to pump blood through the systemic circulation.

To describe the average response to standing over the whole study, each variable was plotted as a function of the minute of standing, and a second-degree polynomial was fitted to all the available measurements pooled together (Figure 5).

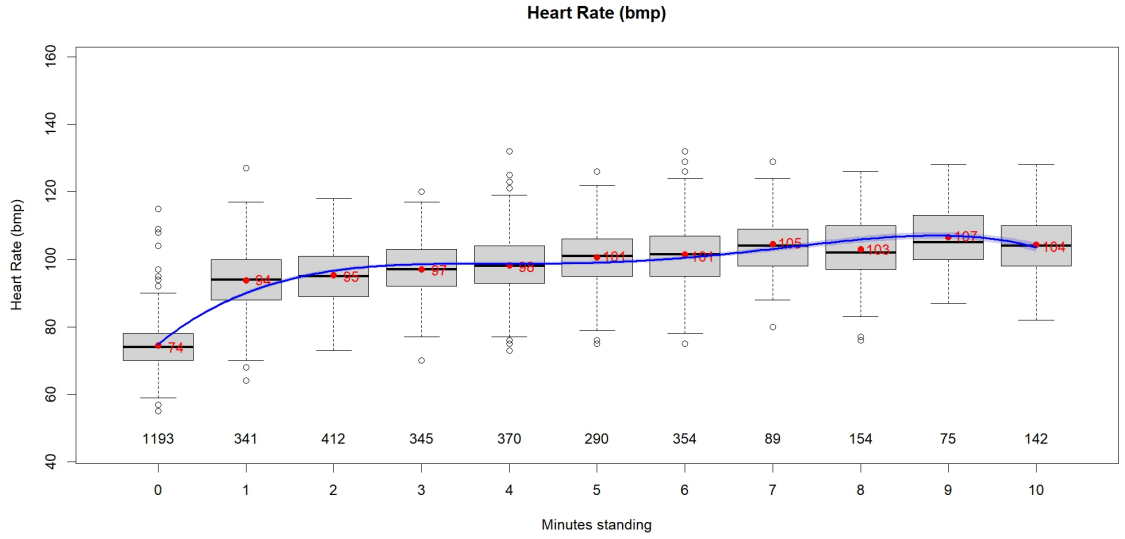


Figure 5: Heart rate during the lean test as a function of the minute of standing (0 = supine). Boxes show the distribution over all tests; red dots and numbers are the means; the numbers at the bottom are the numbers of available measurements. The blue line is a four-degree polynomial fit with its 95% confidence band.

2.2.5 Body composition

Body weight and body composition were measured with a bioelectrical impedance analysis (BIA) scale (Healtheep FG263LB) connected via Bluetooth to a smartphone running the Fitdays application. Measurements were performed in the morning, after voiding, with the subject barefoot and wearing only light clothing. The scale uses a four-electrode, foot-to-foot configuration, with

two electrodes on the front and two on the back of each foot. The scale measures the impedance of the body to a weak alternating current at 50 kHz, and estimates the body compartments from the impedance by means of proprietary equations that also include weight, height, sex, and age [**?**]. The scale measures weight directly, while all the other quantities are derived from the impedance by the proprietary equations of the manufacturer, which are not disclosed. Their absolute values should therefore be interpreted with caution, and in this study they were used only to follow changes over time in the same subject. Measurements were available from 2020-12-29 to 2026-10-24, for a total of 1633 days.

The following variables, exported by the application, were considered: weight, body mass index (BMI), body fat, subcutaneous fat, body water, skeletal muscle, and protein (as percentages of body weight); muscle mass and bone mass (in kg); the visceral fat index; and the basal metabolic rate (BMR, in kcal/day). The masses of skeletal muscle, subcutaneous fat, and body fat were obtained by multiplying the corresponding percentages by the weight.

Since the scale was not used every day, the series were completed by imputation. The days between two measurements were filled by linear interpolation.

2.2.6 Electroencephalography

Resting-state electroencephalography (EEG) was recorded with a consumer-grade wearable headband (Muse S Athena, Interaxon), a type of device whose use for research has been validated against laboratory systems [10]. The headband has four dry electrodes, two frontal (AF7, AF8) and two temporo-parietal, behind the ears (TP9, TP10), with the reference at Fpz. Recordings started on 2025-06-18 and, up to 2026-11-04, were available for 462 days, and were collected once a day, simultaneously with the EEG and HRV recordings, at rest. The measurements were performed in the same room, with the subject laying supine on a bed, in a quiet environment, with the lights off and the eyes closed, after at least five minutes of rest in the same position. The raw signals of the four channels were acquired with the Mind Monitor application and exported in text format, one file per recording [11]. The same files also contain the optical channels used for fNIRS (Section 2.2.7). Only the first recording of each day started before 17:00 was retained, and the first and last 10 s of each recording were removed. The sampling frequency F_s was estimated for each recording from the number of samples and their time stamps.

Mind Monitor marks the samples at which it detects eye blinks and jaw clenches, the main sources of artefacts in frontal and temporal recordings. For each of these events, the samples of all four channels within 0.5 s of the event were removed. Each recording was then divided into consecutive, non-overlapping windows of 10 s, giving a frequency resolution of 0.1 Hz. Windows that contained removed samples were discarded.

For each window and channel, the amplitude spectrum was computed by fast Fourier transform with the R package eegkit [12], and the power spectral density was taken as the square of the amplitude. The power in each of the classical bands, delta (0.5 Hz to 4 Hz), theta (4 Hz to 8 Hz), alpha (8 Hz to 13 Hz), beta (13 Hz to 30 Hz), and gamma (30 Hz to 45 Hz), was obtained by integrating the spectrum over the band with the trapezoidal rule. It was then divided by the

total power between 0.5 and 45 Hz, giving the relative power of the band:

$$P_{c,b}^{\mathrm{rel}} = \frac{\int_b |X_c(f)|^2 \, df}{\int_{0.5}^{45} |X_c(f)|^2 \, df}, \tag{4}$$

where $X_c(f)$ is the amplitude spectrum of channel c and b is the band. Relative powers were preferred to absolute powers because they are independent of the overall amplitude of the signal, which, for dry electrodes, depends on the contact between the electrodes and the skin and may change from one session to another. For each channel and band, the relative powers were averaged over the windows of the recording, giving one value per day.

Besides the 20 channel–band combinations, three indices were computed for each band by summing the relative powers over channels: over all four channels (e.g. `Alpha`), over the left channels AF7 and TP9 (`Alpha_L`), and over the right channels AF8 and TP10 (`Alpha_R`).

2.2.7 Functional near-infrared spectroscopy

Changes in the concentration of oxygenated (HbO_2) and deoxygenated (Hb) haemoglobin in the frontal region were estimated by continuous-wave functional near-infrared spectroscopy (fNIRS), using the optical sensors of a consumer-grade wearable headband (Muse S Athena, Interaxon), the same device used for the EEG recordings. The theory behind the method and its adaptation to this device are described in detail in a previous work [13], to which the reader is referred for the derivation of the equations reported below. Recordings started on 2025-09-19 and, up to 2026-11-04, were available for 379 days, and were collected once a day, simultaneously with the EEG and HRV recordings, at rest. The measurements were performed in the same room, with the subject laying supine on a bed, in a quiet environment, with the lights off and the eyes closed, after at least five minutes of rest in the same position.

The headband has a single optical source at the centre of the forehead, emitting at 660 nm, 730 nm and 850 nm, and two receivers on each side: an inner receiver at about 1 cm and an outer receiver at about 3 cm from the source. Each receiver also records ambient light, with the source switched off. The raw intensities were acquired with the Mind Monitor application and exported in text format, one file per recording, with the 16 optical channels assigned as in Table 1 [11]. Only the first recording of each day started before 17:00 was retained, recordings without full optical data were discarded, and the first and last 10 s of each recording were removed. Each channel was then summarised by its mean intensity over the recording, giving one value per day for each receiver and wavelength.

Table 1: Optical channels of the headband, as numbered in the files exported by Mind Monitor [11], and source–receiver distance L.

Receiver	730 nm	850 nm	660 nm	Ambient	L (cm)
Left outer	1	3	9	11	3
Right outer	2	4	10	12	3
Left inner	5	7	13	15	1
Right inner	6	8	14	16	1

Haemoglobin concentrations were derived from the 730 nm and 850 nm channels using the

modified Beer–Lambert law [14]. Since the intensity emitted by the source is unknown, only changes in attenuation with respect to a reference can be measured. As a reference, the mean of the daily intensities over the whole study, $\bar{I}_{r,\lambda}$, was used for each receiver r and wavelength λ, under the assumption that all the recordings were performed in the same resting conditions. The change in attenuation on day d is then

$$\Delta A_{r,\lambda}(d) = \ln \frac{\bar{I}_{r,\lambda}}{I_{r,\lambda}(d)}. \tag{5}$$

The optical path was taken as $x_{r,\lambda} = L_r \cdot \mathrm{DPF}(\lambda, a)$, where L_r is the source–receiver distance and DPF is the differential pathlength factor of the frontal human head, which depends on the wavelength λ (in nm) and on the age a (in years) of the subject [15]:

$$\mathrm{DPF}(\lambda, a) = 223.3 + 0.05624\, a^{0.8493} - 5.723 \cdot 10^{-7} \lambda^3 + 0.001245\, \lambda^2 - 0.9025\, \lambda. \tag{6}$$

The age was computed as the calendar year of the recording minus the year of birth. This equation was derived for wavelengths between 690 and 832 nm [15], so its value at 850 nm is a small extrapolation. For each receiver and day, the changes in concentration were obtained by solving the linear system

$$\begin{bmatrix} \varepsilon_{\mathrm{Hb}}(\lambda_1) & \varepsilon_{\mathrm{HbO_2}}(\lambda_1) \\ \varepsilon_{\mathrm{Hb}}(\lambda_2) & \varepsilon_{\mathrm{HbO_2}}(\lambda_2) \end{bmatrix} \begin{bmatrix} \Delta[\mathrm{Hb}] \\ \Delta[\mathrm{HbO_2}] \end{bmatrix} = \begin{bmatrix} \Delta A_{r,\lambda_1}/x_{r,\lambda_1} \\ \Delta A_{r,\lambda_2}/x_{r,\lambda_2} \end{bmatrix}, \qquad \lambda_1 = 730\,\mathrm{nm},\ \lambda_2 = 850\,\mathrm{nm}, \tag{7}$$

where $\varepsilon_{\mathrm{Hb}}$ and $\varepsilon_{\mathrm{HbO_2}}$ are the molar extinction coefficients tabulated by Prahl [16]: 1102.2 and 390 at 730 nm, and 691.32 and 1058 at 850 nm, in $\mathrm{cm^{-1}\,M^{-1}}$. These values follow the decadic convention of optical density and were multiplied by $\ln 10$ to be consistent with the natural logarithm used for the attenuation. With paths in centimetres, concentrations are obtained in M.

For each receiver, two derived indices were computed: the change in total haemoglobin, $\Delta[\mathrm{HbTot}] = \Delta[\mathrm{HbO_2}] + \Delta[\mathrm{Hb}]$, a proxy for local blood volume, and the difference $\Delta[\mathrm{HbDiff}] = \Delta[\mathrm{HbO_2}] - \Delta[\mathrm{Hb}]$, an index of oxygenation, whose positive values indicate that the increase in oxygenated blood is greater than the increase in deoxygenated blood, with respect to the reference. Therefore, $\Delta[\mathrm{HbDiff}]$ helps to dentify days on which the increase in concentration of frontal oxygenated blood is not only due to an increase in blood volume.

Frontal indices were obtained by summing the values of the left and right receivers, separately for the inner and the outer receivers. Because the light reaching the inner receivers travels mostly through the scalp and skull, the outer channels are expected to be the more sensitive to cortical haemodynamics. The 660 nm and ambient channels were not used for the concentrations, but their daily means were retained as additional variables.

2.2.8 Blood tests

2.2.9 Drugs and supplements

2.3 Data processing

2.4 Dynamic regression

Measurements repeated on the same subject over time are not independent, and methods that assume independence, such as correlation coefficients, t-tests, and standard regression, are therefore not suited to N-of-1 observational data, although they have often been used in this context [17]. In particular, in the presence of positive autocorrelation, standard regression tends to overestimate the significance of a predictor [18]. Among the methods proposed for these data, dynamic regression is one of the simplest [19, 20]: the outcome is regressed on the predictors and on its own past values, introduced as endogenous covariates, which absorb the autocorrelation of the series. An implementation in R of the pipeline described by McDonald and colleagues [20] was presented in a previous work [21]. In the present implementation of that pipeline, predictors are summarised over time windows of increasing length, to capture delayed and cumulative effects; the number of lags of the outcome is selected automatically for each outcome, and a time term accounts for long-term linear trends; and the significance of the statistical tests on regression coefficients is corrected for multiple comparisons.

Preparation of covariates. For each covariate, the number of days of use and non-use (for non-negative covariates, such as drug doses), the number of available days, and the coefficient of variation were first computed on the raw data. Gaps of up to five consecutive days were then filled by linear interpolation [22], and longer gaps were left missing. Each covariate was detrended by taking the residuals of its linear regression on time, and standardised to zero mean and unit variance. Interactions between covariates were included as the product of the raw daily values of the variables involved, computed before the steps above and then processed like any other covariate. Whenever a product was included, the variables forming it were also included as covariates.

Time windows. Since the effect of a covariate on symptoms may be delayed and may accumulate over several days, each covariate x was summarised over windows of increasing length n. For $n = 0$, the value on the same day as the outcome was used. For $n \geq 1$, the covariate on day t was replaced by its weighted mean over the n preceding days,

$$\tilde{x}_t^{(n)} = \frac{\sum_{d=1}^{n} w_d \, x_{t-d}}{\sum_{d=1}^{n} w_d}, \qquad w_d = \frac{M - d}{M - 1}, \tag{8}$$

where the sums run over the days with available data and $M = 32$. The weights decrease linearly with the distance from the outcome, from 1 on the previous day to 0 for the day M days earlier, so the effect of a covariate is assumed to fade linearly over about one month. Windows extending before the start of the series were not computed. The windows considered varied according to the covariates: for drugs and supplements, as an example, they were $n = 0, 1, 8, 16, 32$ days. After weighted means were computed, the windowed covariates were standardised again.

Standardization was used to make the coefficients of different combinations of covariates and windows comparable. Weighted means were computed with the R package `runner` [23].

Outcome. The outcome was standardised without imputation, so days without a measurement did not contribute to the models. Two versions were analysed: the standardised outcome itself (linear regression), and a binary variable equal to 1 on days in the upper quartile of the outcome and 0 otherwise (logistic regression).

Models. For each covariate, window, and version of the outcome, the following model was fitted:

$$g\left(\mathrm{E}[y_t]\right) = \beta_0 + \beta\,\tilde{x}_t^{(n)} + \gamma\,t + \sum_{l=1}^{p} \phi_l\,y_{t-l}, \tag{9}$$

where g is the identity for the linear models and the logit for the logistic models, t is time in days, and y_{t-l} is the standardised outcome l days earlier, used in both linear and logistic models. Following McDonald et al. [20], the lagged outcomes were included as endogenous covariates, to account for the autocorrelation of the outcome, and the time term accounts for its long-term trend. Hence β measures the association of the covariate with the outcome beyond what is expected from the trend and the recent course of the outcome. The number of lags p was selected separately for each outcome with the Akaike information criterion, among autoregressive models of order 0 to 30 fitted to the outcome after removal of its linear trend, using the function `ar` of R [24].

The models were fitted in two ways. In the marginal analysis, each covariate entered a separate model, so its coefficient also reflects the associations that the covariate shares with other, correlated covariates. In this analysis, products of covariates were treated as composite exposures. In the joint analysis, all the covariates of a predefined group (e.g. meteorological variables) entered the same model, so each coefficient is adjusted for the other covariates of the group. Interactions were interpreted only in this analysis, where the product is estimated together with its main effects.

Multiple comparisons. For each outcome, version of the outcome, type of analysis, and group of covariates, the p-values of all covariates and windows were corrected jointly with the Benjamini–Hochberg procedure [25], and associations with an adjusted $p < 0.05$ were considered significant, which corresponds to a false discovery rate of 5%. For associations with an unadjusted $p < 0.05$, partial regression plots were produced using function `crPlots` from the `car` package [26]. For the joint analysis, the signed $-\log_{10}$ of the adjusted p-values was plotted for each covariate as a function of the window length, positive for positive coefficients and negative for negative ones.

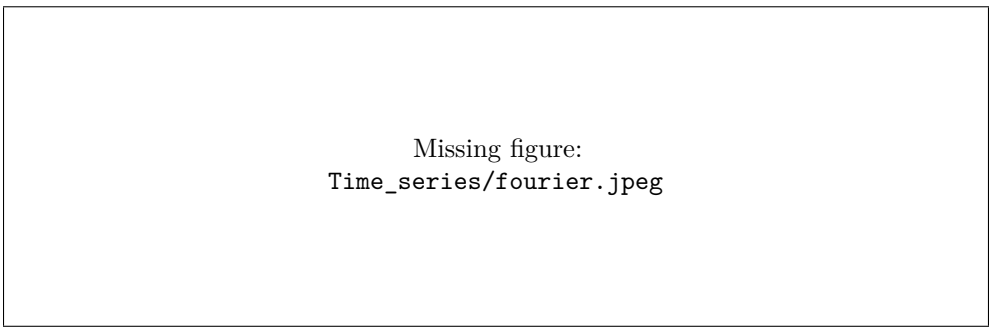


Figure 6: CAPTION.



(a) CAPTION A.



(b) CAPTION B.

Figure 7: CAPTION OF THE PAIR.

Table 2: CAPTION.

Variable	Value	Unit
NAME	0.0	ms

3 Results

4 Discussion

4.1 Limitations

5 Conclusions

Data and code availability

Code: `https://github.com/USERNAME/N1-LoSt`.

Acknowledgements

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A Supplementary material